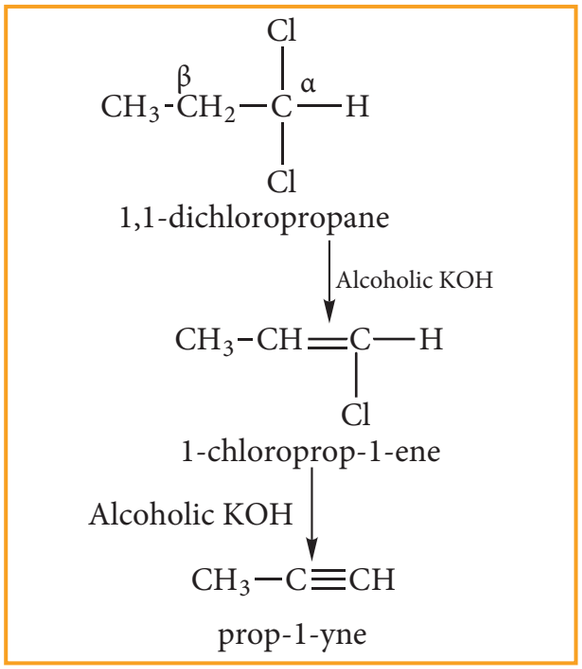
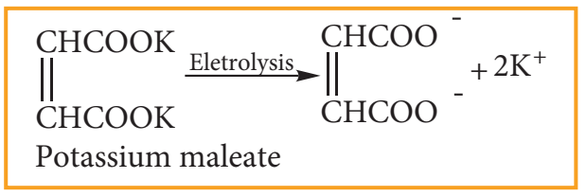
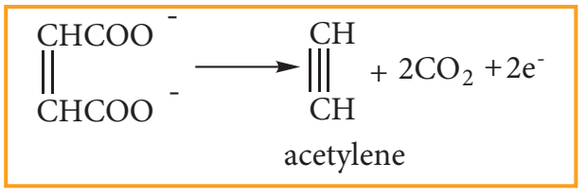



### 3. Preparation of alkynes from electrolysis of salts of unsaturated dicarboxylic acids. (Kolbe's electrolytic method)

Electrolysis of sodium or potassium salt of maleic or fumaric acid yields alkynes.



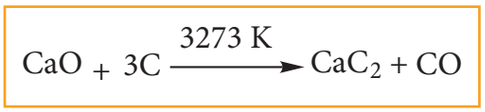




### 4. Industrial prefaration of ethyne:

Ethyne can be manufactured in large scale by action of calcium carbide with water.



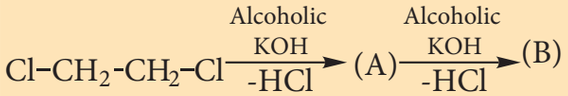

Calcium carbide required for this reaction is prepared by heating quick lime and coke in an electric furance at 3273 K




### Evaluate Yourself

18) Prepare propyne from its corresponding alkene.

19) Write the products A & B for the following reaction.




### 13.4.2. Physical properties of alkynes:

1. The first three members are gases, next eight are liquids and the higher alkynes are solids. They are all colourless and odourless except acetylene which h as garlic odour.

2. They are slightly soluble in water but dissolve readily in organic solvents like benzene, acetone and ethyl alcohol

### 13.4.3. Chemical properities of alkynes

Terminal Alkynes are acidic in nature. It undergoes polymerization and addition reaction.

### 1. Acidic nature of alkynes:

An alkyne shows acidic nature only if it contains terminal hydrogen. This can be explained by considering sp hybrid orbitals of carbon atom in alkynes. The percentage of s-character of sp hybrid orbital (50%) is more than $sp^2$ hybrid orbital of alkene (33%) and $sp^3$ hybrid orbital of alkane (25%). Because of this, Carbon becomes more electronegative facilitating donation of $H^+$



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